

Calc Proofs

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Abstract

Yeah, this is where I'll put my proofs in the future for safekeeping so I can reference them later or smth.

1 Limits

Limit proofs hinge on the following definition of a limit:

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \varepsilon > 0, \\ \exists \delta \text{ s.t. } 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$$

1.1 Addition Rule

Statement:
given that

$$\lim_{x \rightarrow a} f(x) = L, \lim_{x \rightarrow a} g(x) = M$$

, show that

$$\lim_{x \rightarrow a} f(x) + g(x) = L + M$$

Proof. Consider some $\varepsilon > 0$, then by the definition of the limit there exists δ_1 and δ_2 such that:

$$0 < |x - a| < \delta_1 \implies |f(x) - L| < \frac{\varepsilon}{2}$$

$$0 < |x - a| < \delta_2 \implies |g(x) - M| < \frac{\varepsilon}{2}$$

Then let $\delta = \min(\delta_1, \delta_2)$, and we need

$$|f(x) + g(x) - (L + M)| < \varepsilon$$

We have:

$$0 < |x - a| < \delta$$

which implies

$$|f(x) - L| < \frac{\varepsilon}{2}$$

and

$$|g(x) - M| < \frac{\varepsilon}{2}$$

By the triangle inequality,

$$|a + b| \leq |a| + |b|$$

so

$$|f(x) + g(x) - (L + M)| < |f(x) - L| + |g(x) - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

so

$$|f(x) + g(x) - (L + M)| < \varepsilon$$

and so limits follow the addition rule. ■ ■

Do note that I *did* need to consult the textbook for the triangle inequality stuff, but the rest was fine.

1.2 Multiplication Rule

Given that

$$\lim_{x \rightarrow a} f(x) = L$$

and

$$\lim_{x \rightarrow a} g(x) = M$$

prove that

$$\lim_{x \rightarrow a} f(x)g(x) = LM$$

Proof. ■

1.3 Polynomials are continuous

A function, $f(x)$, is continuous if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

From this, let us prove that all polynomials of the form x^n for $n \in \mathbb{N}$ are continuous.

1.4 Rational Powers

Now, since a certain someone asked for it, I'll do *one* more epsilon delta proof.

Prove that

$$f(x) = \sqrt[n]{x}$$

is continuous for all $x > 0$

Proof. Consider some $\varepsilon > 0$, we have to find some δ such that:

$$0 < |x - a| < \delta \implies |\sqrt[n]{x} - \sqrt[n]{a}| < \varepsilon$$

for all $a > 0$.

Which is equivalent to

$$-\delta < x - a < \delta \implies -\varepsilon < \sqrt[n]{x} - \sqrt[n]{a} < \varepsilon$$

Technically $x \neq a$, but we don't need this condition.

Presume $\varepsilon < \sqrt[n]{a}$. This can always be satisfied because $a > 0 \implies \sqrt[n]{a} > 0$.

This implies $0 < \sqrt[n]{a} - \varepsilon < \sqrt[n]{a}$, and we can set $\delta = \min(a - (\sqrt[n]{a} - \varepsilon)^n, (\sqrt[n]{a} + \varepsilon)^n - a)$

$$-(a - (\sqrt[n]{a} - \varepsilon)^n) < x - a < (\sqrt[n]{a} + \varepsilon)^n - a$$

$$(\sqrt[n]{a} - \varepsilon)^n - a < x - a < (\sqrt[n]{a} + \varepsilon)^n - a$$

$$(\sqrt[n]{a} - \varepsilon)^n < x < (\sqrt[n]{a} + \varepsilon)^n$$

Yay! $\sqrt[n]{x}$ is a strictly increasing function, so we can safely:

$$\sqrt[n]{x} - \sqrt[n]{a} < \sqrt[n]{(\sqrt[n]{a} + \varepsilon)^n} - \sqrt[n]{a} \tag{1}$$

$$= \sqrt[n]{a} + \varepsilon - \sqrt[n]{a} \tag{2}$$

$$= \varepsilon \tag{3}$$

and then we also have

$$\sqrt[n]{x} - \sqrt[n]{a} > \sqrt[n]{(\sqrt[n]{a} - \varepsilon)^n} - \sqrt[n]{a} = \sqrt[n]{a} - \varepsilon - \sqrt[n]{a} = -\varepsilon \tag{4}$$

Which therefore implies that

$$-\varepsilon < \sqrt[n]{x} - \sqrt[n]{a} < \varepsilon$$

■

There is some weird domain stuffs, but I think I handled it fine. Figuring out the delta, I obviously needed the textbook for (my brain wasn't big enough), but for the rest I could figure it out...

2 Basic Derivatives

I hate epsilon delta proofs, no more! Derivative proofs are so much more intuitive, its great...

2.1 Power Rule for $n \in \mathbb{N}$

Easy proof with binomial theorem.

Proof. consider $f(x) = x^n$, $n \in \mathbb{N}$.

$$\frac{d}{dx} x^n = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \quad (5)$$

$$= \lim_{h \rightarrow 0} \frac{-x^n + \binom{n}{1}x^{n-1}h + \binom{n}{2}h^2x^{n-2} + \dots + \binom{n-2}{h}n^2 + \binom{n-1}{n}h^{n-1}n + \binom{n}{n}h^n}{h} \quad (6)$$

$$= \lim_{h \rightarrow 0} \frac{-x^n + x^n + nhx^{n-1} + \frac{n(n-1)}{2}h^2x^{n-2} + \dots + \frac{n(n-1)}{2}h^{n-2}n^2 + nh^{n-1}n + h^n}{h} \quad (7)$$

$$= \lim_{h \rightarrow 0} \frac{nhx^{n-1} + \frac{n(n-1)}{2}h^2x^{n-2} + \dots + \frac{n(n-1)}{2}h^{n-2}n^2 + nh^{n-1}n + h^n}{h} \quad (8)$$

$$= \lim_{h \rightarrow 0} \frac{h(nx^{n-1} + \frac{n(n-1)}{2}hn^{n-2} + \dots + \frac{n(n-1)}{2}h^{n-3}n^2 + nh^{n-2}n + h^{n-1})}{h} \quad (9)$$

$$= \lim_{h \rightarrow 0} nx^{n-1} + \frac{n(n-1)}{2}hn^{n-2} + \dots + \frac{n(n-1)}{2}h^{n-3}n^2 + nh^{n-2}n + h^{n-1} \quad (10)$$

$$= nx^{n-1} \quad (11)$$

■

Light work, no help needed.

2.2 Quotient Rule

Quotient rule is pretty easy too. No help needed.

Proof.

$$\frac{d}{dx} \frac{f(x)}{g(x)} = \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} \quad (12)$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x+h)}{h \cdot g(x)g(x+h)} \quad (13)$$

$$= \lim_{h \rightarrow 0} \frac{1}{g(x)g(x+h)} \cdot \frac{f(x+h)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{h} \quad (14)$$

$$= \lim_{h \rightarrow 0} \frac{1}{g(x)g(x+h)} \cdot \frac{g(x)(f(x+h) - f(x)) + f(x)(g(x) - g(x+h))}{h} \quad (15)$$

$$= \lim_{h \rightarrow 0} \frac{1}{g(x)g(x+h)} \cdot \left(g(x) \frac{f(x+h) - f(x)}{h} - f(x) \frac{g(x+h) - g(x)}{h} \right) \quad (16)$$

$$= \lim_{h \rightarrow 0} \frac{1}{g(x)g(x+h)} \cdot (g(x)f'(x) - f(x)g'(x)) \quad (17)$$

$$= \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2} \quad (18)$$

■